

CS & IT ENGINEERING



THEORY OF COMPUTATION

Regular Language

Lecture No. - 04



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Recap of Previous Lecture



Topic

?????



Regular Language Detection
closure properties of Regular language

Pumping lemma
Mehy & Moore m/c.

Topics to be Covered



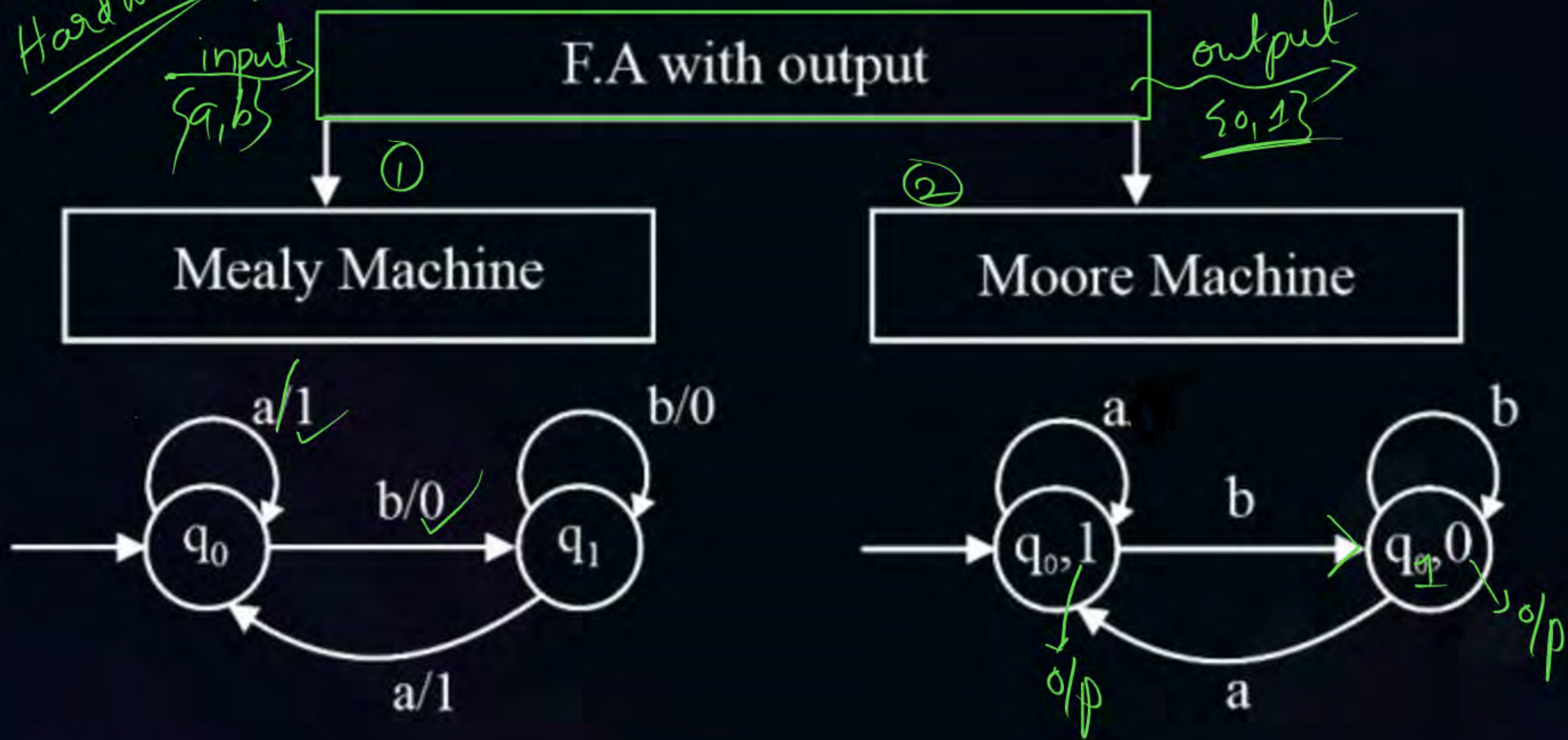
Topic

Regular Language Detection

Topic

Pumping Lemma

Digital Hardware Design



Formal Definition: $\{Q, \Sigma, q_0, \Delta, \delta, \lambda\}$

Q : finite number of states ✓
 Σ : input alphabet ✓
 q_0 : initial state ✓
 Δ : Output Alphabet ✓
 δ : transition function: $Q \times \Sigma \rightarrow Q$

$\boxed{x F x}$

λ : Output function

Mealy m/c: $Q \times \Sigma \rightarrow \Delta$

Mooore m/c: $Q \rightarrow \Delta$

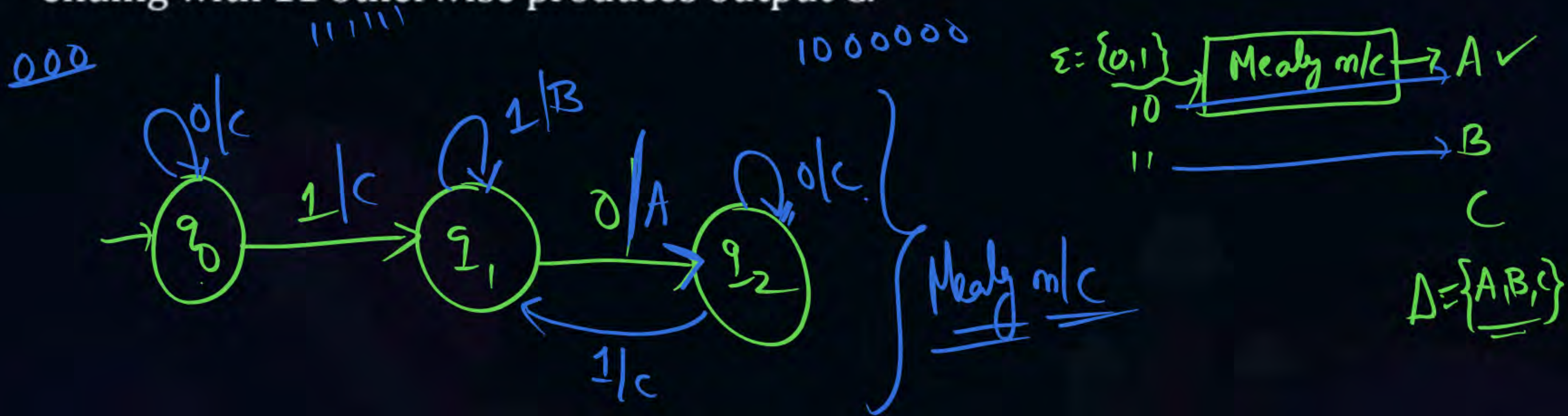
- Mealy Machine:

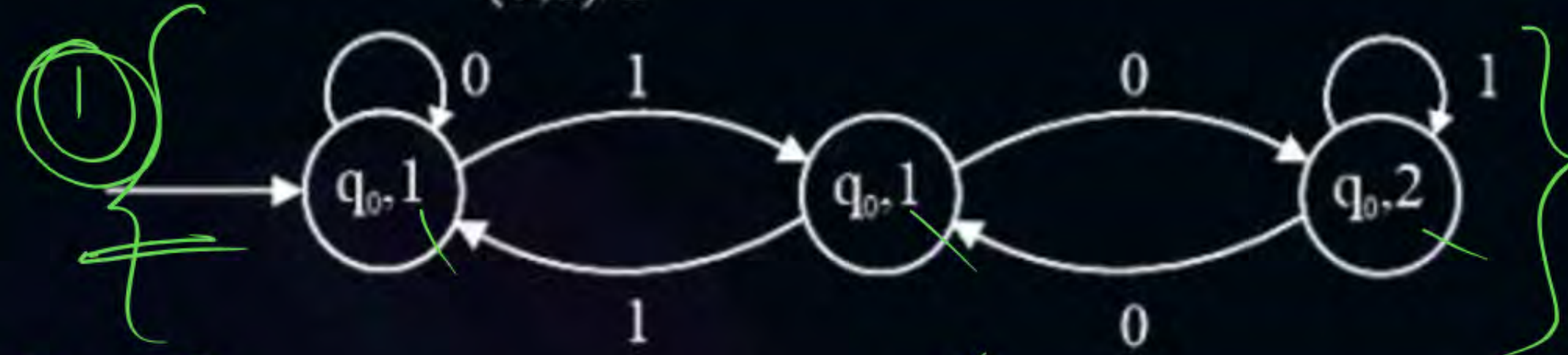
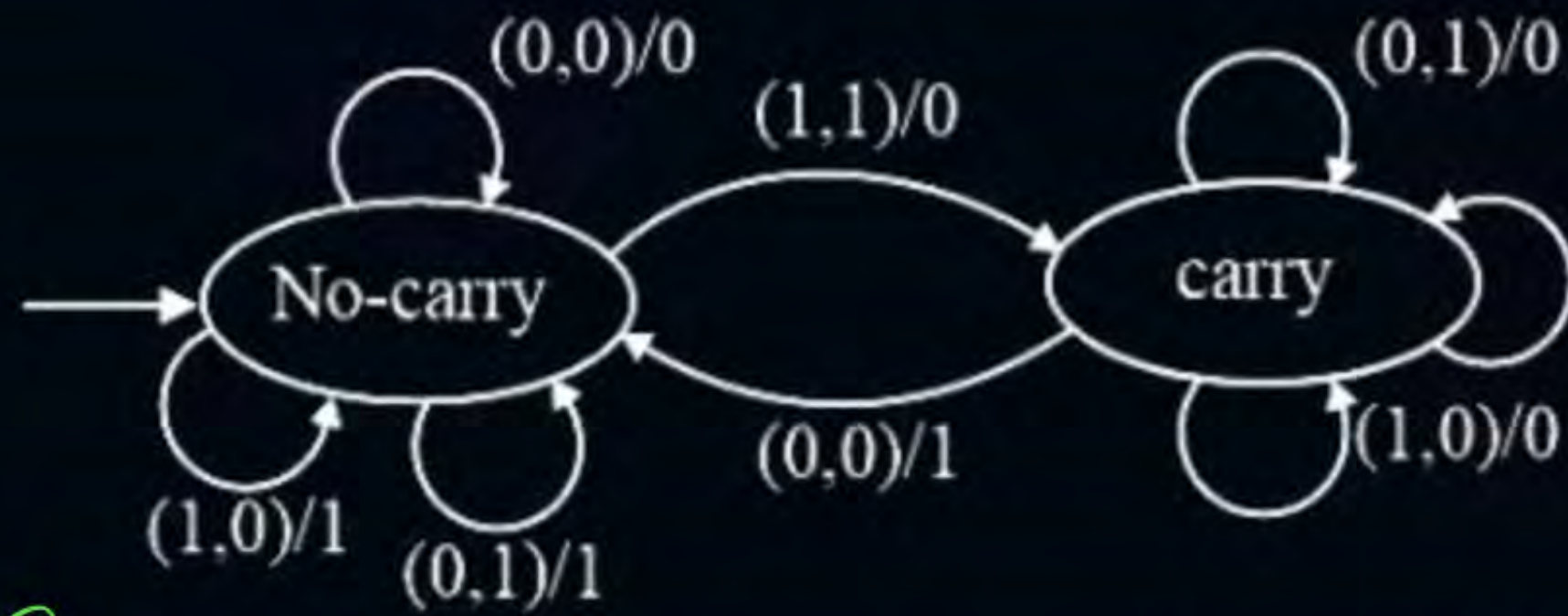
- It is a mathematical model in which output is associated
- with transition.

- Moore Machine:

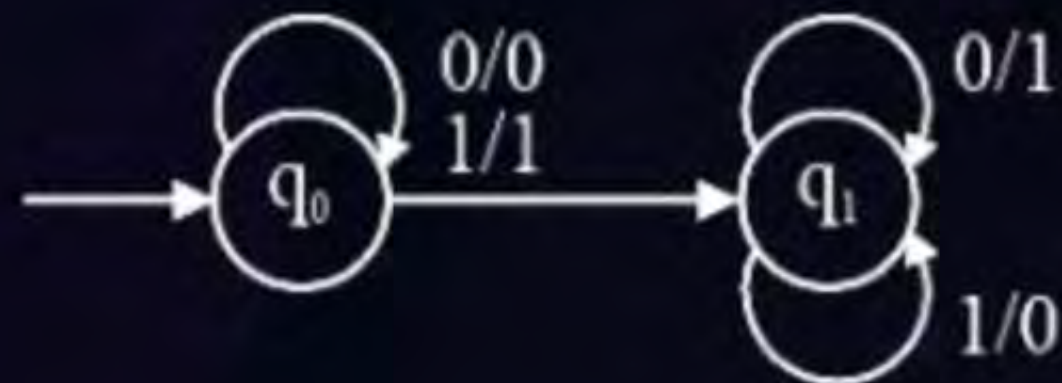
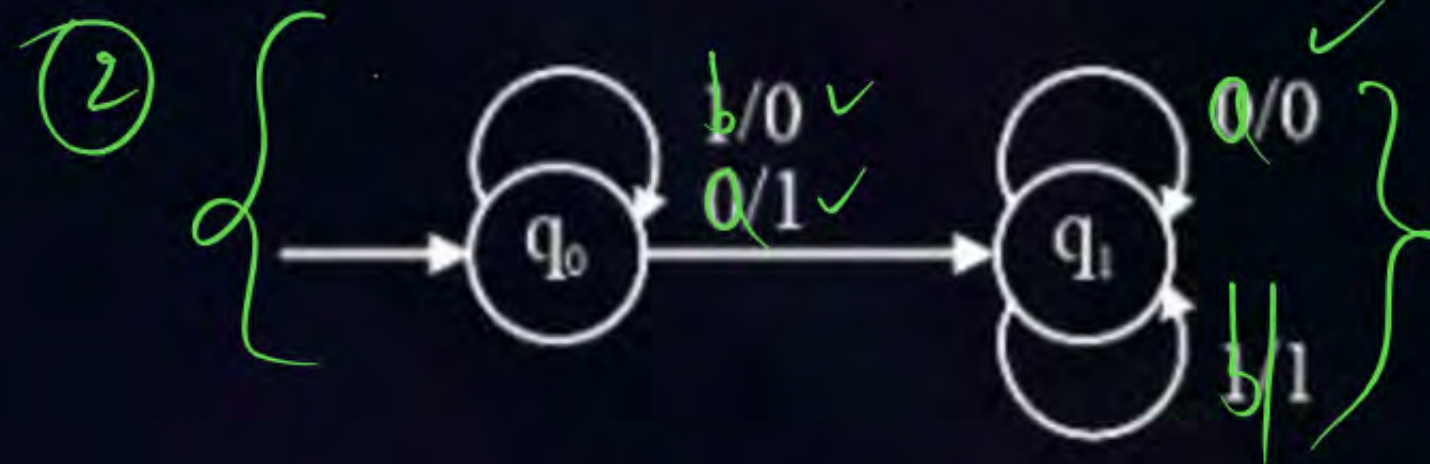
- It is a mathematical model in which output is associated
- with state.

#Q. Construct mealy machine that takes all strings of 0's and 1's as input and produces A as output if input ending with 10(or) produces B as output if input ending with 11 otherwise produces output C.



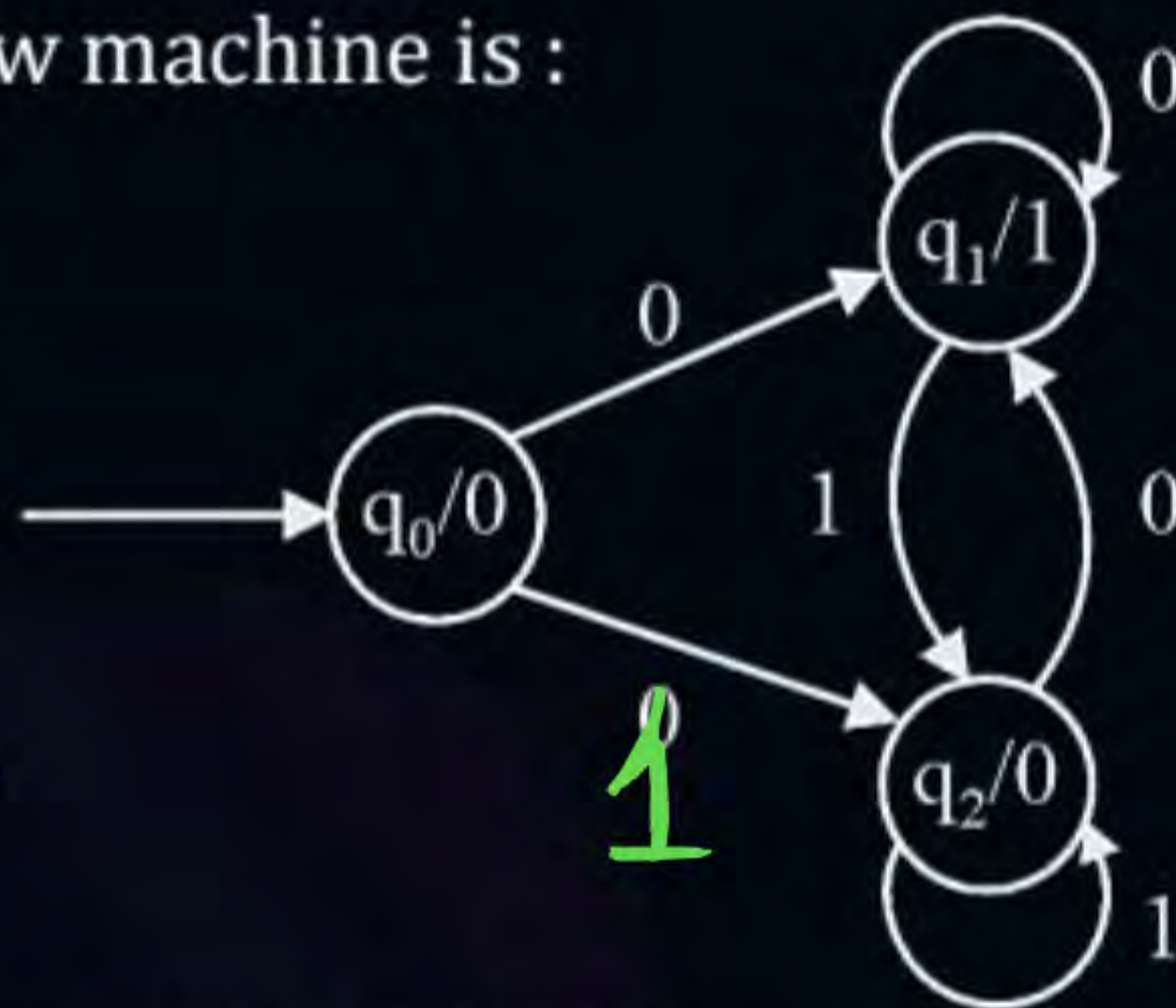


Moore m/c



[MCQ]

#Q. The below machine is :



0 0 1 1 0 1
1 1 0 0 1 0

1's Complement

A

A Mealy machine to find 2's complement of a number

B

A Moore machine to find 2's complement of a number

C

A Mealy machine to find 1's complement of a number

D

A Moore machine to find 1's complement of a number

#Q. A finite state machine with the following state table has a single input x and a single output z .

	Present state $x = 1$	Next state, z $x = 0$
A	D, 0	B, 0
B	B, 1	C, 1
C	B, 2	D, 1
D	B, 1	C, 0

If the initial state is unknown, then the shortest input sequence to reach the final state C is :

A

0 1

B

1 0

C

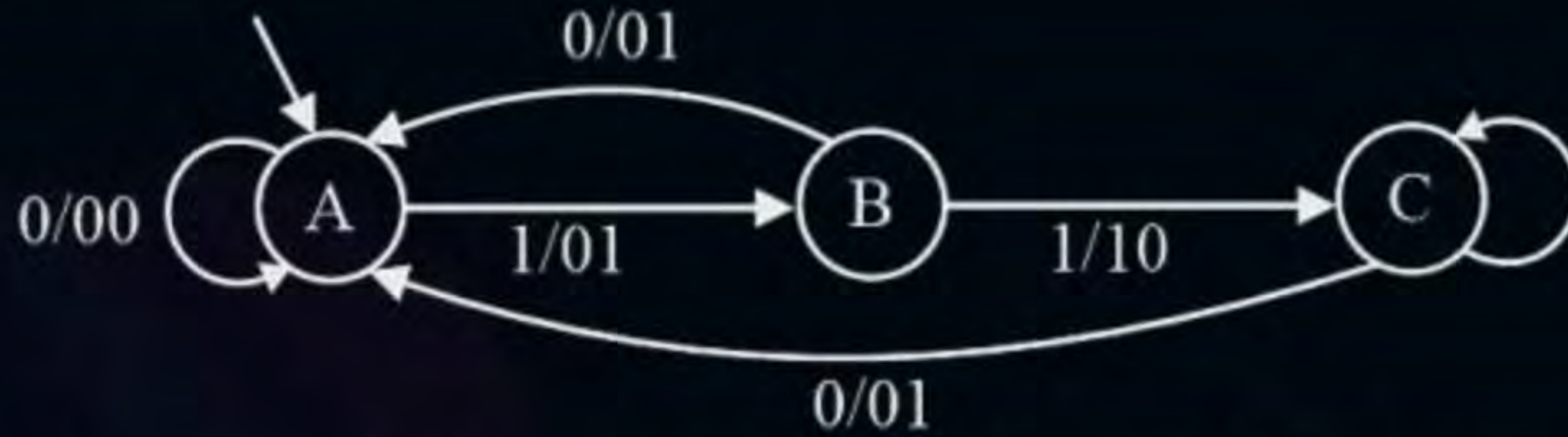
1 0 1

D

1 1 0

[MCQ]

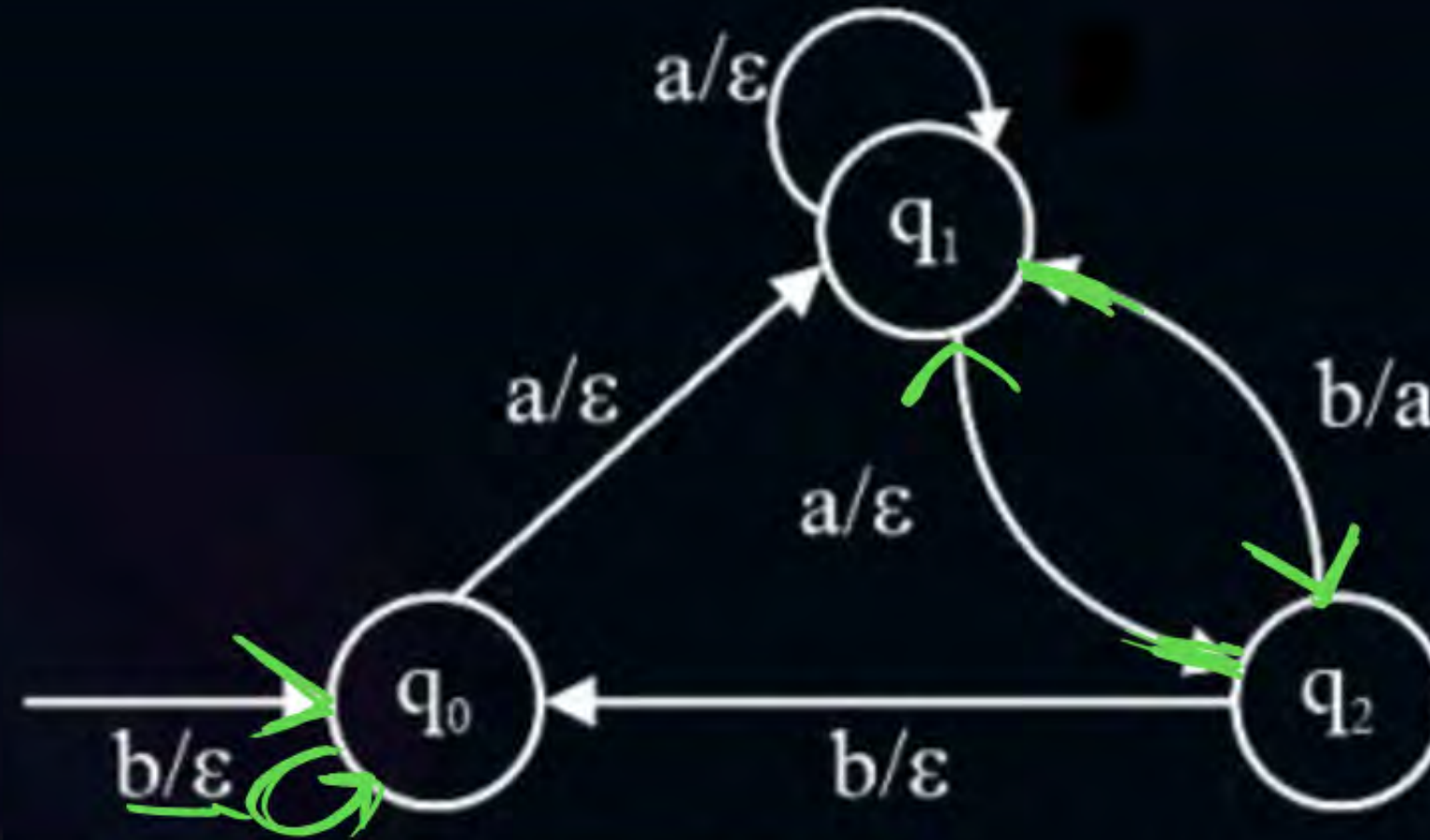
#Q. The Finite state machine described by the following state diagram with A as starting state, where an arc label is x/y and x stands for 1-bit input and y stands for 2-bit output.



- A** Outputs the sum of the present and the previous bits of the input
- B** Outputs a "01" whenever the input sequence contain "11"
- C** Outputs a "00" whenever the input sequence contains "10"
- D** None of the above

[MCQ]

#Q. Consider the following finite state transducer where the label on an edge x/t denotes if the input is x , follow the arrow and emit t



Mealy m/c

For the input, aabbbaaabbbaabaabb the output is :

aaaa
a a a a

☒ A

aaaa

☐ B

aaaaaaaaa

☐ C

ab ab ab ab

☐ D

abbbabbbbababb

THANK - YOU